GRAVITYSORT · REGULATORY AND STANDARDS REGISTER

GravitySort regulatory and standards register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First issue (kit 1.8 pipeline): mercury rules, machinery safety standards, water and smelting, with applicability and status; sources checked on 2026-10-25

GravitySort regulatory and standards register

Which standards and regulations bear on GravitySort, where, and what each needs. The design's main regulatory driver is positive: mercury bans and the Minamata Convention create the need for a mercury-free concentration step. The machinery safety standards are used as design practice for guards and reach distances; no certification is claimed. Not legal advice; anything marked "to assess" needs a qualified reviewer, working with the partner organization, before any product claim. Method: STANDARDS section 23. Mining-area standards are researched here first; no other mining repo in the portfolio has a register yet.

Table 1. Regulatory and standards register.

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G1	Minamata Convention on Mercury, Article 7 and Annex C (artisanal and small-scale gold mining)	International (parties)	Parties with ASGM must reduce and where feasible eliminate mercury use; Annex C names whole-ore amalgamation and open burning of amalgam for elimination and calls for national action plans [1]	GravitySort is a concentration step that replaces amalgamation entirely (PR-001)	Design lead	Met by design (GVS-CAL-001 R1 design review)
G2	Colombia, Law 1658 of 2013 (mercury ban in mining from July 2018)	Colombia	Bans mercury in mining [2]	First partner country (decided by Amish on 2026-10-02); the machine uses no mercury	Partner organization	Met by design; the partner's own permits and mining title to assess with the partner
G3	Regulation (EU) 2017/852 on mercury	European Union	Bans mercury exports and mercury use in ASGM amalgamation in the EU [3]	Does not regulate equipment used in Colombia; relevant to EU-funded programmes that would support GravitySort abroad	Project lead	Does not apply (context only)
G4	ISO 12100:2010, safety of machinery, risk assessment and risk reduction	International	General principles for machinery design and the risk assessment method [4]	Used as the method behind the hazard analysis in GVS-RSK-001; no conformity claimed	Design lead	Applies as practice; formal assessment to assess before any field use
G5	ISO 13857:2019, safety distances to prevent hazard zones being reached	International	Reach distances through and over guard openings [5]	Belt, chain and table head guards are perforated sheet; hole size against distance to the pulleys and belts is not yet checked	Design lead	To assess (perforation size and guard distances, before TRL 4)
G6	ISO 14120:2015, guards: general requirements for fixed and movable guards	International	Design and construction of guards [6]	Lid guard with brake interlock, belt and chain guards (PR-012)	Design lead	Applies as practice; met by design review in part (interlocked lid); guard fixings and strength to assess
G7		European Union	Essential health and safety requirements and CE marking for	Only if units were ever sold or supplied in the EU; an open	Project lead	Does not apply now; to assess if units are ever

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
	Regulation (EU) 2023/1230 on machinery		machinery placed on the EU market [7]	design built by local workshops abroad is not placed on the EU market by the project		supplied in the EU
G8	Colombian machinery safety and occupational health rules for mining	Colombia	Workplace safety at mining sites	No Colombian machinery standard identified for this register	Partner organization	To assess with the partner
G9	Water use and discharge permits for mining sites	Colombia (and any partner country)	Abstraction, settling ponds and discharge of process water	GravitySort recirculates about 1.19 m3/h through a settling pond (PR-008)	Partner organization	To assess per site with the partner
G10	Borax (disodium tetraborate decahydrate) hazard classification	European Union CLP (harmonised)	Classified reproductive toxicant, Repr. 1B, H360FD [8]	Borax is used at the smelting step outside the machine; handling advice belongs in the partner's smelting procedure	Partner organization	Applies to the smelting step (outside this design); safety note in GVS-PRC-001 and GVS-RSK-001
G11	Battery, charger and electrical safety of the motor option	Per country	Lithium packs and the MotionCore module	Covered in the MotionCore repository (shared lab module); the pack is left to the user	MotionCore lead	To assess in MotionCore

Status: to assess, applies, does not apply, met by design, certification needed later.

Notes

- The national action plan under the Minamata Convention in the partner country decides which practices miners must stop and by when; GravitySort's role is to give a mercury-free step that works, not to certify compliance.
- No device certification is expected at TRL 3. If units were ever supplied beyond the partner's own trial, G4 to G7 would need a formal assessment.

Sources

1. Minamata Convention on Mercury, "Guidance document: developing a national action plan to reduce and, where feasible, eliminate mercury use in artisanal and small-scale gold mining," 2017. https://minamataconvention.org/sites/default/files/documents/forms_and_guidance_document/ASGM_guidance_e_2017.pdf (checked 2026-10-25).

2. Mongabay, "Colombia bans the use of mercury in mining," August 2018. https://news.mongabay.com/2018/08/colombia-bans-the-use-of-mercury-in-mining/ (checked 2026-10-25).
3. Regulation (EU) 2017/852 of the European Parliament and of the Council on mercury. https://eur-lex.europa.eu/eli/reg/2017/852/oj/eng (checked 2026-10-25).
4. ISO 12100:2010, Safety of machinery: General principles for design: Risk assessment and risk reduction. https://www.iso.org/standard/51528.html (checked 2026-10-25).
5. ISO 13857:2019, Safety of machinery: Safety distances to prevent hazard zones being reached by upper and lower limbs. https://www.iso.org/standard/69569.html (checked 2026-10-25).
6. ISO 14120:2015, Safety of machinery: Guards: General requirements for the design and construction of fixed and movable guards. https://www.iso.org/standard/59545.html (checked 2026-10-25).
7. Regulation (EU) 2023/1230 of the European Parliament and of the Council on machinery. https://eur-lex.europa.eu/eli/reg/2023/1230/oj (checked 2026-10-25).
8. Instituto Nacional de Seguridad y Salud en el Trabajo (INSST), "Tetraborato disódico, decahidrato," Infocarquim database. https://www.insst.es/agentes-quimicos-infocarquim/sustancias/tetraborato-disodico-decahidrato (checked 2026-10-25).